

Understanding Clinician Engagement with AI in Healthcare: Cognitive and System-Level Perspectives

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Abstract

The effective and safe integration of artificial intelligence (AI) into clinical practice depends not only on technical performance but also on the cognitive and organisational processes that shape how clinicians engage with these tools. Although these factors are increasingly recognised as critical for adoption and use, comparatively little attention has been paid to the psychological mechanisms that underpin clinician–AI interaction in real clinical environments. This PhD addresses this gap by synthesising established psychological literatures and examining their relevance across diverse clinical settings. The thesis begins with a narrative review of three core domains in decision-making psychology - trust, advice-taking, and group-based decision-making - to identify mechanisms through which clinicians interpret, evaluate, and use AI. This synthesis underpins empirical work at Oxford University Hospitals NHS Foundation Trust, combining evaluation of Ambient Voice Technologies with ethnographic case studies in dermatology and intensive care.

CCS Concepts

• **CSS CONCEPTS**; • **Field studies**; • **User studies**; • **Collaborative interaction**;

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1 INTRODUCTION

1.1 Background: AI in clinical practice

Artificial intelligence (AI) systems are increasingly promoted as tools that could enhance clinical decision-making across specialties [6, 7]. Yet these promises sit alongside a persistent gap between increasingly strong technical performance of AI and limited evidence that they have the impact on the diagnostic, workload, or efficiency challenges they are often presented as addressing [1]. Despite the rapid expansion of AI development, only a small subset of tools has achieved sustained clinical adoption. In practice, many clinical environments remain in exploratory or pre-implementation phases,

where systems are assessed, trialled, adapted, or ultimately set aside before reaching routine use [2, 5]. In agreement with Hua et al. [3], Rosenbacke et al [4], and Younis et al. [7], this thesis argues that clinicians' engagement with AI cannot be understood without considering both the psychological mechanisms that shape how clinicians evaluate AI systems and the sociotechnical environments in which this occurs. Developing a clearer account of these intertwined processes is essential for identifying the conditions required to support their effective and safe use in clinical decision-making.

1.2 Objectives and contributions of this PhD

As AI tools become more present in healthcare settings, understanding their real-world influence on clinical judgement becomes increasingly important. To address this, the PhD pursues three linked aims:

- Identifying psychological mechanisms relevant to clinician–AI interaction through a narrative synthesis of trust, advice-taking, and group-based decision-making literatures
- Examining how these mechanisms manifest in practical clinical work across different contexts at Oxford University Hospitals NHS Foundation Trust, drawing on ongoing service evaluations and clinical initiatives and, under appropriate research governance, extending them through ethnographic observations, interviews, and focus groups
- Integrating cognitive and systems insights to propose a cross-case framework that explains variations in clinician engagement and provides guidance for implementation of AI systems

The contribution of this thesis is both conceptual and applied. Conceptually, it aims to advance a cognitive–systems framework that integrates psychological mechanisms with sociotechnical influences, addressing a gap in existing research that tends to separate individual cognition from clinical context. Empirically, it hopes to demonstrate how clinicians' reasoning both shapes and is shaped by the organisational environments into which AI tools are introduced. This offers insights into why AI systems vary in their uptake and how they might be implemented more effectively and safely in practice.

2 METHODS

2.1 Overview of methodological approach

This DPhil adopts a qualitative design with four methods: ethnographic observations, semi-structured interviews, questionnaires, and focus groups. Triangulation across data from these methods



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will enable comparison between anticipated, described, and observed behaviour.

2.2 Case site selection

While much of the literature examines contexts where AI is already in use, many clinical environments remain at early stages of AI adoption. Clinicians therefore engage with AI not only through direct use but also through evaluation of emerging tools. The empirical work draws on service evaluations conducted across several departments at Oxford University Hospitals (OUH), which together span different positions along the AI implementation continuum:

- **ICU – pre-implementation:** The ICU is currently exploring several AI-enabled decision-support tools, none of which have yet been integrated into routine practice. The site therefore provides insight into the cognitive and organisational conditions shaping the feasibility of AI integration in a complex, team-based clinical environment. Observations at this site were initially undertaken as part of service development and evaluation activity.
- **Dermatology – evaluation and non-adoption:** Dermatology has already undergone an evaluation of a diagnostic AI tool used in other NHS trusts, ultimately deciding not to adopt it. This stage provides insight into how clinicians interpret risks and evaluate clinical value, and how cognitive mechanisms interact with system-level constraints during decisions on whether to adopt a specific AI tool. Observations and interviews at this site were initially undertaken as part of service development and evaluation activity.
- **Ambient Voice Technologies (AVT) – early adoption:** AVT tools were piloted across six departments at OUH as part of a service evaluation, providing insight into how clinicians build early experience and adjust workflows.

Ethnography is central to the thesis because it captures work-as-done rather than work-as-imagined [6]. While controlled psychological studies offer valuable insights into cognitive mechanisms, they often abstract away from the social and organisational complexities that shape clinical reasoning. Ethnography makes these complexities visible.

Semi-structured interviews will be conducted with clinicians and non-clinical staff. Interviews provide access to cognitive constructs that are not always visible during observation and allow comparison between anticipated use in pre-implementation sites and experienced use in early-implementation sites. They provide insight into when and why clinicians accept, question, or reject AI recommendations, complementing what is captured in ethnography.

Focus groups will be used to explore group-level reasoning, collective attitudes, and shared norms around AI. The group format also captures how clinicians articulate concerns, challenge assumptions, and co-construct ideas about AI when interacting with colleagues.

3 NARRATIVE REVIEW

Psychological perspectives are increasingly recognised as central to understanding how clinicians evaluate and integrate such tools. While there has already been valuable integration of psychological

perspectives into this work, commentaries consistently emphasise the need for deeper engagement with psychological theory as AI becomes more embedded in clinical decision-making [5, 7].

Two challenges motivate the present review. First, much of the existing work focuses on specific technologies or specific clinical contexts. While useful, these can make the conclusions highly context dependent. Second, as AI systems and their roles in health-care decision-making are rapidly evolving, findings from individual cases can rapidly become outdated. To address these issues, this review draws on established psychological theory to identify mechanisms that apply across settings and technology types. It examines three core strands of psychological research - trust, advice-taking, and group-based decision-making - and considers their implications for how clinicians engage with AI tools in practice. The three strands were selected because prior work demonstrates their relevance to medical decision-making, and because scoping discussions with clinicians and AI developers confirmed their practical importance. These domains are distinct but connected: relying on AI, taking advice from AI, and incorporating AI into team reasoning can be viewed as points along a continuum of human-AI interaction. Synthesising them provides the conceptual grounding for the empirical chapters, which examine how these mechanisms surface in real-world clinical contexts.

4 DERMATOLOGY AND ICU

4.1 Background and aim

This thesis draws on two in-depth ethnographic case studies to examine how clinicians evaluate, make sense of, and engage with emerging AI tools. Although both settings are at pre-implementation stages, they sit at different points in the AI adoption trajectory and represent distinct cognitive and organisational environments. OUH's dermatology department has already reviewed and decided not to adopt a diagnostic AI tool used in other NHS trusts, providing a context where clinicians have had direct experiences of evaluating AI within an outpatient environment. By contrast, the ICU is in an exploratory phase considering several decision-support tools within a fast-paced, urgent environment. These dynamics prompt clinicians to articulate what they would require from AI and where they foresee opportunities or tensions.

4.2 Methods

Both sites are studied through ethnography, semi-structured interviews, and focus groups. The development of the protocols was done in close collaboration with consultants at both ICU and dermatology, drawing upon elements of co-production and ensuring the studies address real clinical needs.

4.3 Governance and ethics

The dermatology and ICU components of this work were initiated as formally registered and approved Quality Improvement (QI) projects within Oxford University Hospitals NHS Foundation Trust, with the primary aim of informing local service development. In accordance with NHS Health Research Authority (HRA) guidance, these activities were classified as service evaluation and did not require NHS Research Ethics Committee (REC) review. The analytical insights presented here emerged from this evaluation work. As the

programme has evolved and identified themes with potential relevance beyond the local context, prospective follow-on research to further develop and examine these findings is currently undergoing appropriate ethics review.

4.4 Results

4.4.1 Dermatology. So far, 23 hours of ethnographic observation across clinics, multi-disciplinary team meeting, and governance sessions have been completed. One focus group and one interview have been conducted so far, with planned totals of 10 interviews and 1–2 focus groups across clinical and administrative roles. This section presents early, proof-of-concept analysis based on data collected so far. Initial analysis has focused on the skin cancer pathway, not only because it is one of the department's highest-pressure workflows, but also because it is the setting in which an AI triage tool was recently evaluated and ultimately not adopted.

Early analysis indicates that dermatology staff work within a system shaped by capacity constraints, high GP referral volume, and fragmented digital infrastructure. These pressures strongly influenced how they evaluated the AI triage tool. A central tension emerged from the mismatch between GP and secondary-care incentives: GPs often refer urgently “just in case”, increasing sensitivity at the expense of specificity, while secondary care must manage this influx under strict 2WW targets. As a result, workload, rather than diagnostic uncertainty, was the dominant bottleneck. This framed expectations about what an AI tool should achieve.

Local attempts to manage this tension, such as the advice-and-guidance (A&G) service and educational initiatives like Skin Club, help to reduce unnecessary referrals but also add substantial work for the department. Clinicians therefore saw AI's potential value primarily in reducing referral volume and administrative burden, rather than in changing diagnostic decision-making. In this sense, the AI triage tool spoke to a real clinical need.

Professional norms and clinical values also shaped evaluations of the tool. Staff emphasised full-body, tactile examination as integral to good dermatology practice, creating a fundamental misalignment with lesion-specific, image-based triage. As one participant noted, “we would check everything, not just what the GP refers for”, illustrating how embodied diagnostic work clashes with systems that narrow the clinical focus. Cognitive mechanisms such as risk perception, trust calibration, and perceived accountability were prominent: clinicians worried about missed cancers and expressed reluctance to assume responsibility for patients they had not physically examined.

Organisational attitudes toward innovation were positive in principle, but only when tools align with workflow and clinical values. The decision not to adopt the triage tool appeared to be driven less by concerns about algorithmic accuracy and more by misalignment with workflow and clinical values.

Overall, these early findings demonstrate how the decision not to adopt the AI triage tool arose from the interaction of:

- Meso-level system pressures (referral volume, 2WW targets, digital infrastructure),
- Micro-level norms and practices (full body examination values, organisational attitudes) and

- Cognitive mechanisms (trust, uncertainty management, risk perception).

4.4.2 Intensive Care Unit. I have completed 40 hours of ethnographic observations in ICU across ward rounds, handovers, and informal decision-making contexts. Initial ICU analysis has begun. It is at an earlier stage than the dermatology work, and due to space constraints only a brief overview of emerging patterns is provided here. Decision-making in the ICU is highly distributed and collaborative: consultants, resident doctors, nurses, pharmacists, and external specialties all contribute to multi-layered clinical judgements. Ward rounds function as the central moment of shared interpretation, where physiological data, test results, and clinical histories are synthesised into a collective picture of the patient. Many decisions unfold dynamically as information becomes available, and although handover structures information transfer, it also relies heavily on tacit knowledge and varies across individuals. These patterns suggest that advice-taking and distributed trust processes will be particularly salient in ICU, where responsibility and interpretation are shared across professional roles.

Navigating uncertainty is a central part of ICU work. Diagnoses are frequently provisional, clinical trajectories can shift rapidly, and clinicians must anticipate deterioration with incomplete evidence. Observations show continuous uncertainty management, including balancing risks, maintaining multiple possible explanations, revisiting earlier assumptions, and drawing on both formal data and contextual cues such as family perspectives and previous clinical behaviour.

Existing technologies further shape these dynamics in both enabling and constraining ways. Standardised review forms, checklists, and monitoring systems structure information flow, while workarounds such as whiteboards and WhatsApp compensate for gaps in integration. Clinicians already rely extensively on predictive scores and monitoring technologies, shaping expectations that any AI tool must integrate into existing workflows. These dynamics suggest that trust calibration, distributed advice-taking, and collective sense-making will be central mechanisms shaping AI integration in ICU. Together, these early findings position ICU as a meaningful counterpoint to dermatology highlighting how team-based cognition, real-time data environments, and high-urgency decisions shape expectations for AI.

5 AMBIENT VOICE TECHNOLOGIES (AVT)

5.1 Background and aims

Ambient Voice Technologies (AVT) use speech recognition to produce clinical documentation, reducing documentation burden and increasing time for patient care. While not decision-making tools, they influence information capture, workflow, and cognitive load in ways that affect clinical decision-making. The AVT pilot described here was commissioned and conducted by TheHill, the digital innovation team at Oxford University Hospitals NHS Foundation Trust (OUH), as a service evaluation to inform procurement and implementation decisions. The evaluation generated data on clinicians' perceived benefits and risks prior to use, how expectations shifted with hands-on experience, and how psychological and system-level factors shaped adoption within local clinical contexts.

5.2 Methods

The service evaluation involved testing AVT solutions across Audiology, Community Paediatrics, Ear Nose and Throat (ENT), Oncology pre-hysteroscopy telephone clinics, Critical Care (ward rounds and inpatient nursing), and OCDEM diabetes and endocrine services. As part of the operational pilot, each department participated in a structured 6-week trial period. Evaluation methods included pre-implementation, in-pilot, and post-implementation surveys, alongside qualitative interviews conducted to inform implementation decisions.

5.3 Results

Surveys were obtained from 62 pre-implementation participants, 435 in-pilot AI encounters, and 42 post-implementation participants with 27 paired responses, across a diverse set of clinical environments. This variation supports examination of how context shapes AVT use. Open-text survey responses and 23 transcribed qualitative interviews (conducted by TheHill) provide additional depth for thematic analysis. The evaluation generated insights into how clinicians formed and revised trust during deployment. The present analysis draws on survey data but does not include the interview transcripts.

5.3.1 Pre-pilot trust. Before using AVTs, clinicians showed moderate and cautious trust across the three domains. Ability-based trust, confidence in accurate transcription, was the weakest dimension (mean = 3.62 on the five-point Likert scale). Free-text comments reflected concerns about hallucinations, uncommon terminology, and the capacity of AVTs to cope with complex clinical encounters. Integrity-based trust was somewhat higher (mean = 3.83), grounded in familiarity with NHS data governance, although several clinicians still requested clearer information about data storage and use in open-text fields. Benevolence-based trust, that the tool was designed with clinicians' needs in mind, was the strongest dimension (mean = 3.89). Clinicians were generally willing to believe that AVTs were intended to reduce administrative burden but hoped the AVT developers would continue meaningful collaboration with clinicians.

5.3.2 Post-pilot trust. After hands-on use, mean trust scores increased across all three dimensions. Ability-based trust showed the largest improvement (mean rising from 3.62 to 4.11), with clinicians describing the systems as more accurate than expected. Benevolence also increased (from 3.89 to 4.14), reflecting perceived reductions in documentation burden and improved ability to focus on patient interaction. Integrity-based trust remained relatively stable (3.83 to 3.98), with qualitative comments suggesting that use of the system did not substantially change clinicians' understanding of data flows, and several continued to express a desire for clearer information about this.

5.3.3 Interpretation. Findings from this analysis suggest a shift from tentative, expectation-based trust before the pilot to higher levels of experience-based trust afterwards. Accuracy of the AVTs remained imperfect, but higher post-implementation levels suggest clinicians generally found errors manageable and context-dependent rather than grounds to refuse the tool. These findings

illustrate how trust interacts with workflow, local norms, and perceived risk, integrating quantitative survey measures with qualitative context.

6 CONCLUSION

As AI becomes increasingly embedded in healthcare, understanding its real-world influence on clinical judgement is essential. This PhD argues that such understanding requires attention to both cognitive mechanisms and the systems in which they operate. By integrating psychological theory with in-situ empirical research across diverse clinical settings, the thesis aims to provide a nuanced account of clinician engagement with AI and to support the development of AI systems that are not only technically robust but also cognitively and organisationally viable.

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